

Determining the Crystalline Structure of Metals through X-Ray Diffraction

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Abstract

This experiment set out to to measure X-ray diffraction patterns of various solid samples in order to describe their crystal structure at the atomic/molecular level. An already calibrated Siemen !!!!!MODEL NAME!!!! X-Ray diffractometer was used to send X-rays into Copper (Cu), Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples named A and B.

Any crystalline solid consists of a regular 3-dimensional packing of cells, the shape of which corresponds to one of fourteen Bravais lattices. These 3D patterns create multiple parallel planar layers within the crystal of various orientations. X-Rays travelling through these planes will diffract off the atoms and constructively interfere at the detector, creating a peak in photon count in the observed diffraction pattern. From that peak, Bragg's law $n\lambda = 2d \sin \theta$ (where $n = 1$ for a first-order plane) can be used to deduce the inter-planar distance d of the diffraction plane of the crystal. Using this inter-planar distance and the corresponding Miller indices, the Bravais lattice can be determined and the lattice parameter(s) calculated. The Cu and Ni samples were found to have a Bravais lattice of [...], with parameters [...] respectively. The alloy sample of these two was found to have [...]. The Sn and Pb samples were found to have a lattice of [...] and parameters [...] respectively, and their alloy [...], [...]. Finally, the mystery samples A and B were determined to be [...] and [...], with lattices [...] and parameters [...], respectively.

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1 Introduction

2 Methods

A pre-calibrated Siemens diffractometer was used to produce X-Rays, send them at the samples, and detect the diffractions. The diffractometer consisted primarily of an X-ray tube, a sample mounting plate, and a photon detector. The X-ray tube used Copper K_α radiation and was cooled using water. Two sets of filters of 0.6 mm were present on the X-ray tube and the detector, to prevent stray rays from leaving the tube and entering the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. The mounting plate kept the sample in place with a spring-loaded mechanism. The samples were centered by aligning the middle of the sample with a red line marked on the mounting plate. A computer was connected to the diffractometer, with which a .dql measuring parameter file was given to the diffractometer and data was obtained from the detector as .RAW files. The measuring parameters were an initial angle of $2\theta_0 = 5.00^\circ$, angle increments of $\Delta\theta = 0.05^\circ$, with 2.00 s at each step, with a final angle of $2\theta_f = 110.00^\circ$. The total measuring time is roughly 1 h 05 m using the aforementioned parameters. Data acquisition was simple: Mount the desired sample onto the mounting plate, centering it with the red line. Then, on the computer, open the software program [!!!NAME!!!], select the desired parameter file, name the trial, and execute the job. After the job finishes, convert the .RAW file to .UXD.

3 Results

4 Discussion

5 Conclusion